Public 3-torsion companion-module filter

Run `public3-20261022T0132Z`; source freeze `36563882e86dacbbcea0d2e473aca236a116ec97`. Host `a81fcf6dc864`, Debian 13 x86_64, managed four-CPU virtual host, L0. Workloads use public synthetic fixtures or public-parameter checks. No wall-clock or full-cost improvement is admitted.

Raw outputs: run directory. Commands, binary/input hashes, seeds, resource limits and statuses are in execution receipts and the versioned protocols in the study root. No independent-host replay is claimed.

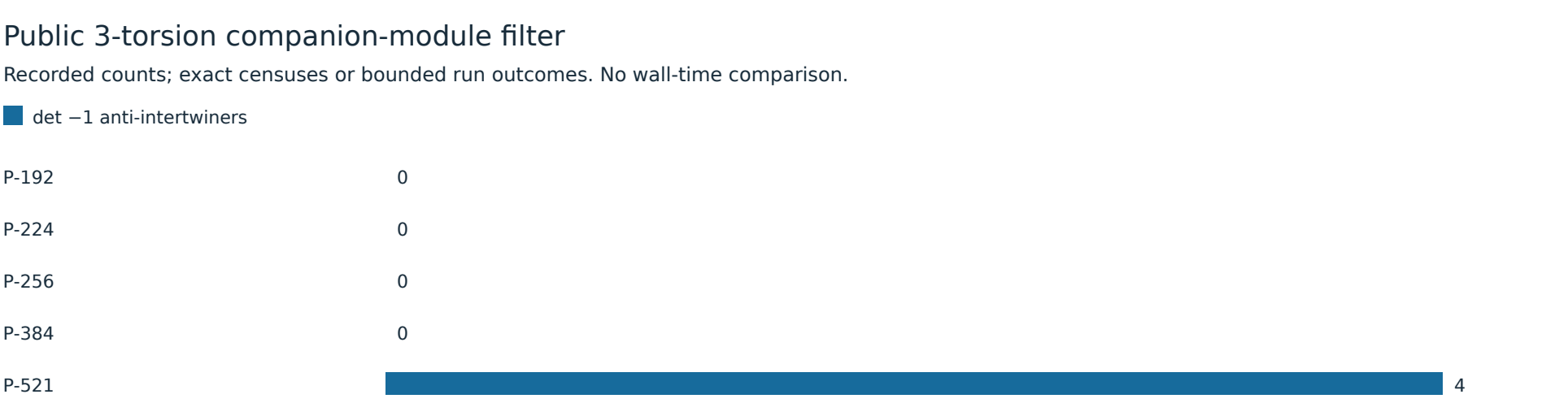
Recorded observations

Metric	Count / recorded value
NIST cases	5
NIST positive module filters	1
geometric constructions certified	0
parameter cases	15
toy cross check mismatches	0
toy cross checks	10

Vector figures and editable data accompany this report on the following pages.

PARI enumerates det(M)=-1 matrices over F3 with MA=-AM for the Frobenius companion matrix. All ten toy counts equal the earlier E11 counts. Among five public NIST P-curve order pairs, only P-521 passes this necessary module filter, with four matrices; the other four have none. This is a public-parameter calculation, not a NIST DLP. The actual torsion/Weil-pairing bridge, principally polarized quotient, genus-two model and divisor maps are not constructed or certified. A positive count is a candidate for further construction, not a speedup or novelty result.

The chart has no statistical interval: census counts are exact on the stated finite sets; solver completion counts describe these bounded processes only. Missing or failed computations stay unresolved. Full original acceptance criteria and proof/measurement gaps remain in the requirement matrix.


Public 3-torsion companion-module filter
Recorded counts; exact censuses or bounded run outcomes. No wall-time comparison.
det −1 anti-intertwiners
P-192
0
P-224
0
P-256
0
P-384
0
P-521
4